A03 · COPY INTO incremental loading

Objective: use COPY INTO to incrementally load files from cloud object storage into Unity Catalog governed tables.
Free Edition: a Unity Catalog volume is the source. Serverless notebook. Fictional orders only.

1. What COPY INTO is

```
COPY INTO target_table            -- must already exist (Delta)
FROM 'path' | (SELECT ... FROM 'path')
FILEFORMAT = CSV | JSON | PARQUET | AVRO | ORC | TEXT | BINARYFILE
[VALIDATE ...] [FILES = (...) | PATTERN = '...'] [FORMAT_OPTIONS (...)] [COPY_OPTIONS (...)]
```

Idempotent: files already loaded into THIS table are skipped, even if edited since.

2. Typed table + COPY INTO from a volume

```
CREATE TABLE orders_copy (order_id INT, customer STRING, amount DOUBLE,
                          order_date DATE, source_file STRING);
COPY INTO orders_copy
FROM (SELECT CAST(order_id AS INT) AS order_id, customer, CAST(amount AS DOUBLE) AS amount,
             CAST(order_date AS DATE) AS order_date, _metadata.file_name AS source_file
      FROM '/Volumes/workspace/bt_copyinto_xxxx/landing/orders')
FILEFORMAT = CSV  FORMAT_OPTIONS ('header' = 'true');
```

3. Rerun = idempotent

```
run 1  orders_day1.csv lands  -> num_inserted_rows 3, total 3
run 2  no new files           -> num_inserted_rows 0, total 3   (no duplicates)
run 3  orders_day2.csv lands  -> num_inserted_rows 2, total 5
```

4. Pick files, test first

```
PATTERN = 'orders_day*.csv'    glob filter
FILES = ('orders_day2.csv')    explicit list, up to 1,000 (not with PATTERN)
VALIDATE 15 ROWS               parse + schema check, writes nothing
```

Options, schema evolution, COPY INTO vs Auto Loader

5. FORMAT_OPTIONS vs COPY_OPTIONS

```
FORMAT_OPTIONS  (the file reader)  'header', 'inferSchema', 'mergeSchema'
                                   (merge source file schemas), 'ignoreCorruptFiles'
COPY_OPTIONS    (the copy itself)  'mergeSchema' = evolve the TARGET table
                                   'force' = 'true' -> idempotency off, reload all
```

force = true reloads every file: expect duplicates.

6. Schemaless table: columns inferred

```
CREATE TABLE IF NOT EXISTS orders_auto;          -- no columns
COPY INTO orders_auto FROM '/Volumes/.../orders' FILEFORMAT = CSV
FORMAT_OPTIONS ('header' = 'true', 'inferSchema' = 'true', 'mergeSchema' = 'true')
COPY_OPTIONS ('mergeSchema' = 'true');           -- 5 rows (new load history)
-- day 3 adds a channel column -> 2 rows, channel added, older rows null, total 7
```

Until the first COPY INTO, a schemaless table can't take INSERT INTO or MERGE INTO.

7. COPY INTO vs Auto Loader

```
COPY INTO     SQL command, load history kept per table, thousands of files,
              easy to reload a chosen subset of files
Auto Loader   cloudFiles streaming source + checkpoint, millions of files,
              fewer discovery operations, richer schema inference + evolution
```

Thousands of files: COPY INTO. Millions: Auto Loader (recap in A02, more in A04).

8. Five exam traps

1) Rerun with no new files: 0 rows. 2) Edited file with the same name: still skipped. 3) force = true: every file loads again, duplicates. 4) New target table: its own load history, so all files load. 5) Evolve the target schema: mergeSchema in COPY_OPTIONS.

Practice challenge

After day 3 lands, rerun the orders_copy COPY INTO: 2 inserted, total 7, no channel column (the SELECT lists columns). Add force = true: 3 + 2 + 2 = 7 inserted, total 14 with duplicates. Two million files a day: Auto Loader.

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